

Contrast (PD / PAX / both) — run report

RUN

012316_pd-visibility_pd-visibility

Status: completed | Started: 2026-09-24T00:23:16.768391+00:00

PURPOSE

Measure PD voltage contrast, PAX power contrast, or both concurrently. Passive mode leaves the drive untouched; active mode drives the selected phase actuator and returns outputs to zero on exit.

REQUIRED OPTICAL SETUP

PD needs a DC-coupled input and measured/provided signed dark baseline. PAX needs the recorded port/wavelength. Both requires light delivered to both sensors (e.g. a beam split), with that arrangement recorded. Active mapping: $\phi_1 = \text{OUT1}$, $\phi_2 = \text{OUT2}$; the other output is zero. Passive PD needs the existing Pyrpl FPGA. Drive must span fringes.

RECORDED BENCH CONTEXT (RECIPE)

Source: pax; mode: active; PD input: in1; PAX location: In front of NPBS 1, C port; requested/declared drive: 0.5 Hz. Active ϕ_1 , sin, 0.36 +/- 0.36 V; outputs zero on exit.

OPERATOR SETUP NOTES

V_{pi} expectation only; verify channel and measured driver gain before calibration.

DATA

85 recorded samples; acquisition: 85

Source: data.csv; full acquisition settings: recipe.json; status: run.json

RAW DOP

Mean 0.9354; range 0.8855 to 0.9811. 0 samples outside 0–1. Telemetry is not a new calibration.

PAX contrast

SOURCE

pax

SAMPLES

85

RECORDED DURATION S

12.115877003991045

EXTERNAL FREQUENCY HZ

0.46566128730773926

POWER MIN W

0.0006929609

POWER MAX W

0.0007436275

POWER MEAN W

0.00071810328

VISIBILITY

0.0352686963085599

SAMPLE INTERVAL MEDIAN S

0.14422298799763666

SAMPLE INTERVAL MAX S

0.14585403600358404

SAMPLES PER DRIVE CYCLE

14.890023274480972

PAX contrast

STATUS

observed PAX total-power contrast

DEFINITION

$(P_{\max} - P_{\min}) / (P_{\max} + P_{\min})$, extrema of recorded PAX total optical power in watts

INTERPRETATION

Uses instrument-reported total power, with no additional optical-background subtraction. No PD voltage zero or polarity is applied. Finite sampling may miss narrow extrema; full fringe coverage and suitable drive speed must be checked. This is not a noise-corrected fit.

MODE

active

Contrast drive

MODE

active

FREQUENCY HZ

0.46566128730773926

AXIS

phi1

OUTPUT

out1

WAVEFORM

sin

ASG WAVEFORM

sin

OFFSET V

0.3599853515625

AMPLITUDE V

0.3599853515625

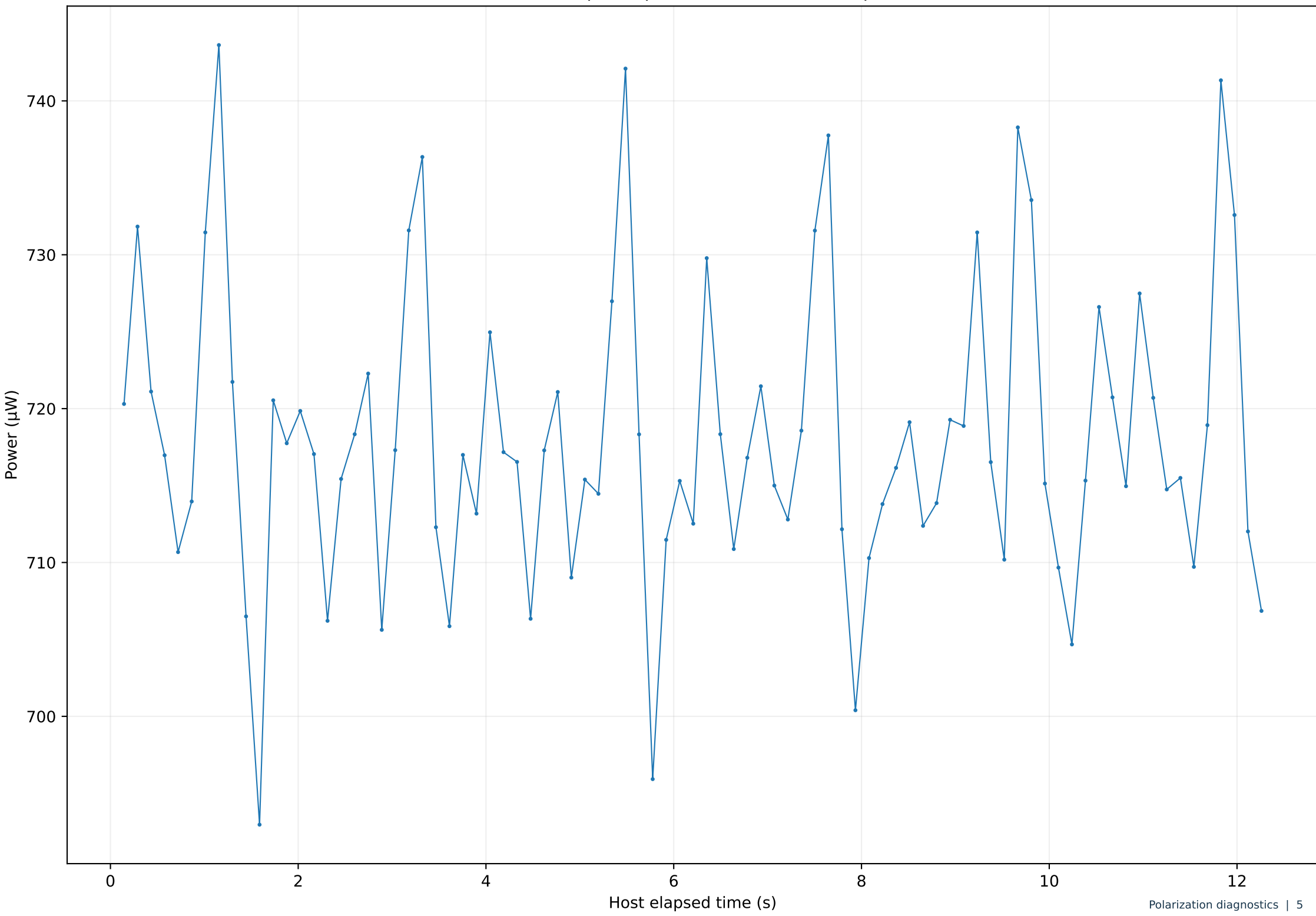
REQUESTED FREQUENCY HZ

0.5

STARTED AT

2026-09-24T00:23:29.513319+00:00

PAX total optical power — recorded samples



Active parameters

Contrast: pax, active. Active drive readback is in drive.json; voltages are RP command volts.

Parameter	Recorded value
label	"pd-visibility"
duration_s	12.0
report	"pdf"
pd_input	"in1"
rp_output_max_voltage	1.0
pax_rotation_velocity_hz	60.0
pax_measurement_mode	5
pax_wavelength_nm	830.0
visibility_frequency_hz	0.5
visibility_source	"pax"
visibility_mode	"active"
visibility_axis	"phi1"
visibility_waveform	"sin"
visibility_amplitude_v	0.36
visibility_offset_v	0.36
visibility_pax_sample_period_s	0.05
visibility_dark_voltage_v	null